

No. of Pages	3
No. of Questions	4
Total Marks	30
Time: 75 minutes	

Department of Computer Science and Engineering
FINAL EXAMINATION Fall 2024
CSE 110: Programming Language I

A

- ✓ Write theory teacher's Name/Initial on top of the answer script in LARGE FONT.
- ✓ Answer all questions. Use **back part** of the answer script for rough work.
- ✓ Answer Question 1 & 2 at the **beginning part** of the answer script.
- ✓ Figure in bracket [] next to each question indicates marks for that question.
- ✓ At the end of exam, put **question paper** inside answer script and **return both**.
- ✓ Understanding the question is part of the exam, **please do not ask questions**.

No washroom breaks

Section: _____ Name: _____

Question 1 [CO3] [10 Points]
[Answer on the answer-script]

Write a Java program that takes a string as input from the user. Assume that the input string consists only of lowercase English letters. A letter may appear one or more times in the string. Your task is to create and print a modified string where the first and last occurrences of any letter that appears more than once in the input string are removed. All other occurrences of such letters, as well as letters that appear only once in the input string, should remain in their original order. If the modified string is empty, it should print "No characters left!".

Sample Input	Sample Output
banana	ba
aabbcc	No characters left!
giggling	ggl

Explanation: For the sample input "banana", among the characters 'b', 'a', and 'n', character 'b' occurs only once. That's why it remains unchanged in the modified string.

Characters 'a' and 'n' occur more than once in the input string. The indexes for 'a' are 1, 3, and 5. In the modified string, the first and last occurrences of 'a' (indexes 1 and 5) are removed.

Similarly, for 'n', the indexes are 2 and 4. Therefore, in the modified string, the first and last occurrences of 'n' (indexes 2 and 4) are removed.

Hence, the modified string is "ba".

Question 2 [CO4] [10 Points]
[Answer on the answer-script]

Write a Java program where an array of integers is given. Now, you have to create two new subarrays: one containing all the odd numbers and the other containing all the even numbers. Finally, print both arrays and the difference between the Largest Even Number and the Smallest Odd Number.

Note: You may use Arrays.toString() to print the subarrays.

[Your code should work for any given integer type array. You can assume given array will have at least one odd integer and one even integer.]

Given Array	Sample Output
int [] array = {5, 3, 8, 6, 7, 2, 1, 4};	Odd Numbers: [5, 3, 7, 1] Even Numbers: [8, 6, 2, 4] Difference between Largest Even Number and Smallest Odd Number: 7
int [] array = {2, 4, 1, 6, 12};	Odd Numbers: [1] Even Numbers: [2, 4, 6, 12] Difference between Largest Even Number and Smallest Odd Number: 11

Question 3 [CO4] [5 Points]

[Answer on the question paper]

The code below is designed to sort an array in increasing order. However, it contains 6 errors. Analyze the code, identify the errors, and correct them. Provide your corrections in the specified format.

1	public class A{
2	public void main(String args []){
3	int array = {20, 25, 10, 8, 3};
4	for (int i = 0; i <= array.length; i++){
5	int min_index = i;
6	for (int j = i+1; j < array.length(); j++){
7	if (array[j] > array[min_index]){
8	min_index = j;
9	}
10	}
11	int temp = array[min_index-1];
12	array[min_index] = array[i];
13	array[i] = temp;
14	}
15	System.out.println("Sorted Array in Increasing Order: ");
16	for (int j = 0; j < array.length; j++){
17	System.out.print(array[j]+" ");
18	}
19	}
20	}

Write your corrections in the table below. For better understanding, one error correction is shown.

Line Number	Corrected Code Line
Line 2	public static void main(String args []){
Line 3	int[] array = {20, 25, 10, 8, 3};
Line 6	for (int j = i+1; j < array.length; j++){
Line 8	int min_index = j;
Line 11	int temp = array[min_index];
Line 17	System.out.print(array[j] + " ");

Question 4 [CO5] [5 Points]
[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. Your answer will not be accepted without the workings. All of the outputs must be in the question paper.

1	public class TracingA {
2	public static void main(String[] args) {
3	int [] b = {20, 5, 15, 30, 13};
4	System.out.println(metA(b, 0));
5	}
6	public static int metA(int[] a, int i) {
7	if (i == a.length - 1) {
8	return a[i];
9	}
10	int x = metA(a, i + 1);
11	System.out.println(a[i] + x);
12	if (a[i] < x) {
13	return a[i];
14	} else {
15	return x;
16	}
17	}
18	}

Output
20
5
15
30
13

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Section

in CAPITAL

Question 1 [CO3] [10 Points]
[Answer on the answer-script]

Write a Java program that takes a string as input from the user. Assume that the input string consists only of lowercase English letters. A letter may appear one or more times in the string. Your task is to create and print a modified string where the first and last occurrences of any letter that appears more than once in the input string are removed. All other occurrences of such letters, as well as letters that appear only once in the input string, should remain in their original order. If the modified string is empty, it should print "No characters left!".

Sample Input	Sample Output
banana	ba
aabbcc	No characters left!
giggling	ggln

Explanation: For the sample input "banana", among the characters 'b', 'a', and 'n', character 'b' occurs only once. That's why it remains unchanged in the modified string.

Characters 'a' and 'n' occur more than once in the input string. The indexes for 'a' are 1, 3, and 5. In the modified string, the first and last occurrences of 'a' (indexes 1 and 5) are removed.

Similarly, for 'n', the indexes are 2 and 4. Therefore, in the modified string, the first and last occurrences of 'n' (indexes 2 and 4) are removed.

Hence, the modified string is "ba".

Question 2 [CO4] [10 Points]
[Answer on the answer-script]

Write a Java program where an array of integers is given. Now, you have to create two new subarrays: one containing all the odd numbers and the other containing all the even numbers. Finally, print both arrays and the difference between the Largest Even Number and the Smallest Odd Number.

Note: You may use Arrays.toString() to print the subarrays.

[Your code should work for any given integer type array. You can assume given array will have at least one odd integer and one even integer.]

Given Array	Sample Output
int [] array = {5, 3, 8, 6, 7, 2, 1, 4};	Odd Numbers: [5, 3, 7, 1] Even Numbers: [8, 6, 2, 4] Difference between Largest Even Number and Smallest Odd Number: 7
int [] array = {2, 4, 1, 6, 12};	Odd Numbers: [1] Even Numbers: [2, 4, 6, 12] Difference between Largest Even Number and Smallest Odd Number: 11

Question 3 [CO4] [5 Points]

[Answer on the question paper]

The code below is designed to sort an array in decreasing order. However, it contains 6 errors. Analyze the code, identify the errors, and correct them. Provide your corrections in the specified format.

1	public class B{
2	public void main(String args []){
3	int [] array = {3, 8, 25, 10, 20};
4	for (int i = 0; i < array.length(); i++){
5	int max_index = i;
6	for (int j = i+1; j <= array.length; j++){
7	if (array[j] < array[max_index]){
8	max_index = j;
9	}
10	}
11	int temp = array[max_index];
12	array[max_index-1] = array[i];
13	array[i] = temp;
14	}
15	System.out.println("Sorted Array in Decreasing Order: ");
16	for (int j = 0; j < array.length; j++){
17	System.out.print(array[j]+" ");
18	}
19	}
20	}

Write your corrections in the table below. For better understanding, one error correction is shown.

Line Number	Corrected Code Line
Line 2	public static void main(String args[])
Line 4	for (int i = 0; i < array.length; i++){
Line 6	max_index = i;
Line 12	array[max_index] =
Line 13	array[max_index-1] = temp;
Line 17	System.out.print(array[max_index-1] + " ");

Question 4 [CO5] [5 Points]

[Answer on the question paper]

Illustrate the outputs of the following statements. Provide your workings on the answer script to verify your outputs. Your answer will not be accepted without the workings. All of the outputs must be in the question paper.

1	public class TracingB {
2	public static void main(String[] args) {
3	int [] b = {9, 15, 11, -2, 7};
4	System.out.println(metA(b, 0));
5	}
6	public static int metA(int[] a, int i) {
7	if (i == a.length - 1) {
8	return a[i];
9	}
10	int x = metA(a, i + 1);
11	System.out.println(a[i] + x);
12	if (a[i] < x) {
13	return a[i];
14	} else {
15	return x;
16	}
17	}
18	}

Output
14
5
16
26
15